Rahul Gupta

Software Development Engineer

+91 89298 08785 Delhi, India
rahulguptaji1019@gmail.com
rxhxul.dev github.com/w-rahul
linkedin.com/in/rahul-gupta-b9a003259

Software engineer with strong fundamentals in backend systems and distributed workflows, and a sharp eye for frontend and product design. Builds with AI agents every day, from planning to review, and holds every change to production standards.

EXPERIENCE

Surepass Technologies

Delhi, India · Oct 2024 - Present

Software Development Engineer II — **Jan 2026 - Present**

SureParser · LLM document platform

- Designed and built SureParser, an LLM platform that classifies, splits and extracts structured data from PDFs and images, served through sync and async APIs with durable background processing.
- Built a workflow engine that lets customers compose document pipelines from extraction, translation, routing, human review and merge steps, with parallel branches and reliable completion callbacks.
- Moved usage-based billing to signed callbacks on live traffic with zero client breakage, and added tenant isolation.
- Cut LLM response latency by 40% and token spend by about 30% through profiling, prompt restructuring and caching.
- Contributed to Console, the multi-tenant web app for the product suite, and mentored a junior frontend engineer.

DPDP Suite · privacy compliance platform

- Built data discovery that scans databases and file stores to find and classify personal data, including Indian identifiers, with long-running scans that checkpoint and resume.
- Built data-subject rights workflows that find a person's data and carry access and erasure requests through verification.
- Strengthened authorization and session security across the console and public consent APIs.
- Designed the console views for scan control and results, keeping complex, long-running state easy to read.

Software Development Engineer I — **Jan 2025 - Dec 2025**

- Extended SureSign, the Python and Java e-signature platform, with certificate-based digital signing from PFX keystores, so every signed document carries a verifiable signature.
- Added signer verification: face match to confirm identity, and geofencing that controls where a document can be signed.
- Built PDF utilities for merging, splitting and other everyday document operations, used across the product line.
- Built identity, land-record and credit verification APIs that unify several third-party providers behind one normalized interface, with timeouts, retries, circuit breakers and failover.

Python Intern — **Oct 2024 - Dec 2024**

- Added email-based authentication, API fallbacks and production fixes on the SureReach backend.

PROJECTS

racetrap · open-source tool that finds race conditions in Postgres-backed APIs — github.com/w-rahul/racetrap

- Built a proxy that speaks the Postgres wire protocol, holds one request before each of its SQL statements while another runs, and reports any result that running the two one at a time cannot produce.
- Found a signup race in the official FastAPI full-stack template (46k stars) in 20 seconds, and reported it with a tested fix.
- On PyPI with a GitHub Action and a pytest plugin. 82 tests, strict typing, CI on three systems and Postgres 14 to 18.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Backend: FastAPI, Flask, Node.js, Express, async SQLAlchemy, Alembic, Pydantic, signed webhooks, idempotent APIs

Data and Infrastructure: PostgreSQL, Redis, MongoDB, S3, Temporal, Docker, Cloudflare Workers, GitHub Actions

LLM Engineering: Structured outputs, tool calling, prompt and response caching, token and latency budgeting, Pydantic AI

Agentic Development: Builds agent harnesses, Pi and Hermes among them, with hooks, MCP servers, scoped permissions and memory. Runs parallel subagents in worktrees and pairs self-written skills with community ones like grill-me and ponytail.

Frontend and Design: React, Next.js, Vue. Designs interfaces end to end, from layout and type to motion, as on rxhxul.dev.

Engineering: System design, concurrency and idempotency, security review (tenant isolation, SSRF, sessions), test-first bug fixes

EDUCATION

B.Tech, Computer Science and Engineering · Jagannath University · CGPA 8/10 — **2019 - 2023**